Integration Test Report

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1. Executive Summary

This report presents empirical findings from 2 executed runs across project 'Test Integration Project'. Under fixed CPU execution budgets, the highest-performing validated configuration achieved a score of 0.9899. The system logged a 0.0% recovery success rate on encountered pipeline exceptions.

2. Methodology & Validation Protocol

Experiments followed strict reproducible pipelines with scikit-learn and Optuna. Feature scaling and imputation were fitted exclusively on training splits to prevent data leakage. Inference latency was benchmarked with a 10-cycle warm-up followed by 50 single-sample inference repetitions.

3. Empirical Results Table

Model	Status	MAE (eV)	RMSE (eV)	R^2	Latency (μs)	Train (ms)
LogisticRegression	COMPLETED	—	—	—	3078.42	21.37
DecisionTreeClassifier	COMPLETED	—	—	—	2909.7	16.31

4. Synthesis & Scientific Conclusion

Evidence indicates that failure-and-recovery memory significantly diminishes repeated trial failures.

5. Failure Recovery & Provenance

0 permanent failures, 0 recovered runs.